复旦数学资料汇编
算子代数

来源：香蕉空间 Solution 资料库

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编者说明

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第一章　算子代数

1.1　试卷: 算子代数

原网页

25 秋 (徐胜芝)

本次考试总时间约为 2.5 小时 (7:40–10:10)

一、 (50pts) Let A be the algebra $\ell^1(\mathbb{Z}^n)$. For $a,b\in\ell^1(\mathbb{Z}^n)$, define $a*b:p\mapsto\sum\limits_{q\in\mathbb{Z}^n}a_qb_{p-q}$, and $a^*:p\mapsto\overline{a_{-p}}$. Let $\chi_q\in A$ be the sequence defined by $(\chi_q)_p=\delta_{p,q}$. For $z\in\mathbb{T}^n$ and $p\in\mathbb{Z}^n$, denote $z^p=z_1^{p_1}z_2^{p_2}\cdots z_n^{p_n}$.

(1) Prove that $\overline{\{\chi_p|p\in\mathbb{Z}^n\}}=A$, $\chi_p*\chi_{p'}=\chi_{p+p'}$ and χ_0 is the unit of A.

(2) For any $a\in A$, define the function $\Phi a:\mathbb{T}^n\to\mathbb{C},z\mapsto\sum\limits_{p\in\mathbb{Z}^n}a_pz^p$. Show that $\Phi(a)$ is a continuous function on $\mathbb{T}^n$.

(3) Prove that Φ is an (involution) algebra morphism and $\|\Phi\|=1$.

(4) Prove that any character $\psi:A\to\mathbb{C}$ is of the form $\psi(a)=\Phi a(z)$ for a unique $z\in\mathbb{T}^n$.

(5) Prove that $a\in A$ is invertible if and only if Φa has no zeros.

二、 (50pts)

(0) Show that every positive functional $\psi:C_0(\mathbb{R})\to\mathbb{C}$ is of the form $\psi(f)=\int_{\mathbb{R}}f(x)\mu(\mathrm{d}(x))$ for a finite Borel measure on $\mathbb{R}$.

(1) Show that the GNS construction of ψ is the multiple representation on $L^2(\mathbb{R},\mu)$ defined by $\pi(f)h=fh$ for $h\in L^2(\mathbb{R},\mu)$.

(2) Find a function $h\in L^2(\mathbb{R},\mu)$ such that $L^2(\mathbb{R},\mu)=\overline{\{\pi(f)h|f\in C_0(\mathbb{R})\}}$ and $\psi(f)=\langle h|\pi(f)h\rangle$.

(3) Show that the multiple representation can be extended to $L^\infty(\mathbb{R},\mu)$. Denote also by $(\pi,L^2(\mathbb{R},\mu))$. Show that the commutant of $\pi(C_0(\mathbb{R}))$ is $\pi(L^\infty(\mathbb{R},\mu))$.

(4) Find the bicommutant of $\pi(C_0(\mathbb{R}))$. Prove that $\pi(L^\infty(\mathbb{R},\mu))$ is a von Neumann algebra over $L^2(\mathbb{R},\mu)$.

1.2　试卷: 算子代数 K 理论

原网页

26 春 (林华新)

注：考试为线上发放试卷，线上提交，需要抄写题目。开卷，可以使用 ai，但不允许和人交流。考试时间 3 小时。

Problem 1. Let A be an amenable C^*-algebra. Prove that the unitization $\widetilde{A}$ is also an amenable C^*-algebra.

Problem 2. Let $\phi : A \to B$ be a completely positive linear map, where A and B are C^*-algebras. Show that

$$\phi(a)^*\phi(a) \leq \|\phi\|\phi(a^*a)$$

for every $a \in A$.

Problem 3. Let A be a C^*-algebra. Suppose that, for every $\varepsilon > 0$ and every finite subset $F \subset A$, there exist an amenable C^*-subalgebra $B \subset A$ and contractive completely positive linear maps

$$\phi : A \to B, \qquad L : B \to A$$

such that

$$\|a - L(\phi(a))\| < \varepsilon$$

for all $a \in F$. Prove that A is amenable.

Problem 4. Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of unital C^*-algebras. Put

$$A = \prod_{n=1}^{\infty} A_n, \qquad B = \bigoplus_{n=1}^{\infty} A_n,$$

and let $\pi : A \to A/B$ be the quotient map. Prove the following statements:

(a) If $p \in A/B$ is a projection, then there is a projection $q \in A$ such that $\pi(q) = p$.

(b) If $u \in U(A/B)$, then there is a unitary $v \in A$ such that $\pi(v) = u$.

Problem 5. Let $n_0 = n_1 = 1$ and $n_{k+1} = n_k + n_{k-1}$ for $k \geq 1$. For $k \geq 1$, set

$$A_k = M_{n_k} \oplus M_{n_{k-1}},$$

where $M_1 = \mathbb{C}$. Thus

$$A_1 = \mathbb{C} \oplus \mathbb{C}, \qquad A_2 = M_2 \oplus \mathbb{C}, \qquad A_3 = M_3 \oplus M_2, \qquad A_4 = M_5 \oplus M_3, \qquad \cdots$$

Define $\phi_k : A_k \to A_{k+1}$ by

$$\phi_k(x, y) = (\mathrm{diag}(x, y), x).$$

Put $A = \varinjlim(A_k, \phi_k)$. Show that, as an additive subgroup of $\mathbb{R}$,

$$K_0(A) = \mathbb{Z} + \mathbb{Z}\left(\frac{1+\sqrt{5}}{2}\right).$$

Problem 6. Let

$$A = \varinjlim_{n\to\infty} (A_n, \phi_n),$$

where each A_n is a simple C^*-algebra. Show that A is simple.

1.3　试卷: 高指标理论

原网页

26 春 (徐胜芝)

本次考试总时间约为 2.5 小时 (8:00–10:30)

一、 (30pts) Suppose that there is an exact sequence of smooth algebras

$$0 \to J \xrightarrow{\varphi} A \xrightarrow{\psi} B \to 0.$$

(0) Give the definition of the exponential and logarithm map to give the 6-term exact sequence of K-groups.

(1) Prove that $K_j(B) = 0$ if B is trivial in multiplication: $b_1 b_2 = 0$ for $b_1, b_2 \in B$.

(2) Prove that A^2 as the closure of span(AA) is an ideal in A, and that A/A^2 is trivially multiplicated. Prove that the inclusion map $\eta : A^2 \to A$ induces an isomorphism on K-groups.

二、 (20pts) For $0 \leqslant t \leqslant 1$, define $(h_t f)(z) = f(tz)$ for $f \in C(\mathbb{D})$, where $\mathbb{D} = \{z \in \mathbb{C} : |z| \leqslant 1\}$.

(1) Prove that $(h_t : C(\mathbb{D}) \to C(\mathbb{D}))_{0 \leqslant t \leqslant 1}$ is a homotopy. Compute the K-groups $K_j(C(\mathbb{D}))$.

(2) Recall that $K_0(C(\mathbb{T})) = \mathbb{Z}[1]$, $K_1(C(\mathbb{T})) = \mathbb{Z}[\zeta]$, where $\zeta(z) = z$. Prove that every continuous function $f : \mathbb{D} \to \mathbb{D}$ admits a fixed point.

三、 (20pts) Recall that the Morita invariance: for $n \in \mathbb{Z}_+$, the map $\theta_1 : x \mapsto x \oplus 0 \oplus \cdots \oplus 0$ and $\theta_n : x \mapsto 0 \oplus \cdots \oplus 0 \oplus x$ from a smooth algebra B to the matrix algebra $B(n)$ induce the same isomorphism on K-groups.

(1) Let B be a closed ideal of a unital smooth algebra D. Prove that the automorphism $ad_b : x \mapsto bxb^{-1}$ for $b \in D^\times$ induces the identical map on K-groups.

(2) Let $\varphi, \psi : A \to B$ be two morphisms between smooth algebras such that $\psi(x) = u\varphi(x)v$ and $\varphi(x) = v\psi(x)u$ for some $\{u, v\} \subseteq D$ with $uvu = u$ and $vuv = v$. Prove that φ and ψ induce the same morphism on K-groups.

四、 (30pts) Let A be a C^*-algebra and H_A be the canonical Hilbert A-module.

(1) Prove that $K_A(H_A) = A \otimes K(l^2)$ and the morphism $\gamma : A \to A \otimes K(l^2), a \mapsto a \otimes e$ induces the same isomorphism on K-groups for any projection e on l^2 of rank 1.

(2) Prove that any adjointable isometry operator $V : X \to H_A$ between Hilbert A-modules induces a morphism $ad_V : T \mapsto VTV^*$ from $K_A(X)$ to $K_A(H_A)$ which induces a morphism on K-groups, denoted by γ_X, independent on V.

(3) Give the definition of (adjointable) Fredholm operators $T : X \to Y$ between Hilbert A-modules and construct its index if $X \oplus Y \oplus Z \cong H_A$ for some Hilbert A-module Z.